FARMARE — Whitepaper

Automated on-chain yield, engineered for the worst regime

Version 1.0 — July 2026. No token. No DAO. Non-custodial vault on Base/Arbitrum.

Abstract

FARMARE is a rules-based, automated yield strategy delivered as a self-custodial ERC-4626 vault on Ethereum L2s. It combines a diversified stablecoin core with a hedged, concentrated liquidity-provision engine, under a risk framework selected on a *maximin* criterion — every design decision is chosen for its worst-case market regime, not its best. Depositors hold vault shares priced at daily NAV and may exit at any time; no party, including the operator, can move deposited funds to a third address. There is no governance token and no DAO: the operator sets the rules, but every rule change is bound by a 48-hour on-chain timelock and hard-coded ceilings, and the depositor's right to redeem is unconditional. **All performance to date is simulated (paper trading on live data); this document is informational and is not an offer of any financial product.**

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1. The problem

On-chain yield is abundant but hostile to passive capital. Stablecoin rates compress the moment capital arrives; concentrated liquidity positions earn rich fees but bleed value in any trending market ("gamma bleed"); incentive farms carry exploit and rug risk. Most retail and even institutional capital either sits in the lowest-yield venues or takes uncompensated directional and smart-contract risk chasing headline APYs. What is missing is not another high-APY venue — it is disciplined, automated *management* of the trade-off between fee income, impermanent loss, and tail risk.

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2. Strategy

FARMARE runs a **core–satellite** allocation, validated in Phase 0 as a static **Prudent 80/20** profile.

Core (≈80%) — stablecoin yield. Capital rotates across whitelisted blue-chip protocols (Aave, Morpho, Fluid, Pendle, Compound, Curve, Aerodrome, Uniswap), diversified over two venues with a hard cap of ~50% per protocol. When a fixed rate beats the best floating rate by ≥1 percentage point, the core **locks that rate** via Pendle Principal Tokens, converting a variable yield into a bond-like fixed return immune to APY compression.

Engine (≈20%) — hedged market-making. Concentrated ETH/USDC liquidity on Uniswap v3 with a volatility-adaptive range, paired with a conditional short-perpetual hedge. The hedge neutralizes the position's directional exposure, turning the LP from a bet on price into a **carry trade on trading fees**. When funding clearly pays more than expected LP fees, the sleeve switches to a delta-neutral funding-capture position; when neither is viable (hyper-volatility or a parabolic trend), a viability guard **parks** the sleeve in the stable core until the economics return.

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3. Where the yield comes from (and why it persists)

1. **Lending / LP spreads** — money-market and AMM equilibrium rates
(5–12% on-chain), paid by persistent on-chain credit demand.

2. **Fixed-vs-floating premium** — PT buyers are paid 2–6pp to absorb rate
compression risk that most capital refuses to hold.

3. **LP fees net of hedged risk** — concentrated liquidity earns outsized
fees; the adaptive range and delta hedge remove most directional and gamma cost that passive LPs systematically mismanage.

4. **Perpetual funding** — longs pay shorts most of the time; captured only when it clearly beats the alternatives.

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4. Evidence

All results are reproducible from the project repository.

Test	Data	Result
Real bear semester (ETH –44%)	Real pool volumes, prices, funding	Hedged LP **+26% vs hold**, –5% max drawdown
Three regimes (bear / flat / bull)	Real + synthetic	Prudent 80/20 never worse than –2pp vs any alternative
Three chains (Base, Solana, BSC)	Real data, same strategy	Hedge thesis confirmed on all three
Decade 2016–2026	Coinbase prices, era-anchored economics	Guarded portfolio positive **every calendar year** (worst +3.7%)
Fixed-rate (PT) vs floating	147 days of Pendle rates	PT ladder **+11.5% vs +5.6%**
Ideas rejected by the data	maximin across regimes	6, incl. trend filters and vol regime-switching

Outcome range by regime (net of modeled costs): portfolio +7.5% (worst tested regime) / +16.3% (sideways) / +10% (bull); engine-only 2023–2026 validation +13% to +22% per year. The worst-case figure is the design floor, not the expectation.

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5. Protocol architecture

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    depositors ■■deposit/withdraw■■■  FarmVault (ERC-4626, on Base)
                                      · shares priced at daily NAV
                                      · venue whitelist + 48h timelock
                                      · per-venue caps · loss circuit breaker
                                      · entry/mgmt/performance fees (HWM)
                                             ■ allocate / deallocate (bounded)
                               ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
                          Venue adapters (vault ■ venue only)
                          Aave v3 · Morpho/ERC-4626 · (Pendle, Uniswap v3)
                                             ▲
                               off-chain KEEPER executes the public algorithm
```

The **keeper** is an off-chain process that signs only `allocate` / `deallocate` / `accrueFees`, exclusively between whitelisted adapters and within caps and a per-operation loss bound. It has no path to withdraw funds. Adapters can only move assets vault → venue → vault. Value (NAV) is read live from the adapters, so any loss marks down immediately and fairly.

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6. Governance — sole operator, no token

FARMARE has **no DAO and no governance token**, by explicit design. The operator (a hardware-wallet owner) is the sole decision-maker for venues, caps, fees and keeper appointment. Depositor protection is structural, not a matter of trust:

- **48-hour timelock** on every rule change — announce-then-apply, so anyone who disagrees can exit *before* it takes effect.
- **Hard-coded ceilings** the operator cannot exceed (management fee ≤5%, performance fee ≤30%).
- **Unconditional exit** — `redeem()` at NAV works at any time, even while new

deposits are paused; withdrawals auto-unwind venue positions.

- **Minimal keeper** — the automation can rebalance but never withdraw.

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7. Fees

Share-based, identical to a classic fund: a management fee accrued pro-rata on AUM and a performance fee charged only on gains **above the high-water mark**, both minted to the operator as share dilution (never taken from principal). Defaults: 2%/year management, 20% performance. All fee changes pass the 48h timelock and the hard ceilings above.

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8. Risk framework

Automatic guards, checked hourly and acted on within 6 hours:

- **Depeg** — any underlying below $0.99 excludes the pool; if held, an immediate emergency exit fires.
- **TVL collapse** — a –30% drop from the 7-day high excludes / exits the pool.
- **Diversification** — the two core slots never share a protocol.
- **Viability guard** — the engine parks when fees stop covering volatility (decade-tested through 2017, 2020 and 2022).
- **Circuit breakers** — a keeper operation reverting on >0.5% single-op loss, plus a rolling 2%/24h cumulative-loss cap.
- **Auditability** — every decision is journaled and git-committed with timestamps; a public dashboard rebuilds from that history.

Quantified worst cases (from the production-code stress suite): a single-candle –35% ETH gap ≈ –2.4% at portfolio level; an underlying depeg while held costs ~0.23% of the affected slot plus drift until the next run; a total data-API outage aborts the run with state intact and no loss.

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9. Security

An internal adversarial review fixed five findings — a first-deposit inflation attack, a reentrancy vector, orphaned funds on venue removal, an unbounded cumulative-loss path, and a UI decimals bug — each with a regression test. The contract suite passes 20/20 (unit, security, fuzz, and Base-mainnet fork tests against real Aave and Morpho). This internal review is **not a substitute for a professional external audit**, which is a prerequisite before third-party funds, alongside a public testnet with a bug bounty and staged deposit caps.

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10. Roadmap

- **Phase 0 — Research & backtest.** Complete: strategy validated across regimes, chains and a decade of data.
- **Phase 1 — Live paper trading.** Active: the strategy runs on live data in a virtual portfolio, every decision published to an auditable track record.
- **Phase 2 — On-chain vault.** In progress: ERC-4626 vault + Aave/Morpho adapters built and fork-tested; deposit/withdraw UI live. Remaining: Pendle and Uniswap-v3 engine adapters, external audit, testnet, then a small pilot.
- **Phase 3 — Scale.** Gradual capital increase and the full hedged engine

once the forward test and audit gates are cleared.

11. Legal & disclaimers

FARMARE does not currently custody or accept third-party funds. All performance shown is **simulated** (paper trading on live market data) and is not a forecast. On-chain deposits, once live, carry real smart-contract and DeFi risk, including total loss. Nothing herein is investment advice or an offer or solicitation of any financial product. Accepting third-party funds with management is a regulated activity (MiCA in the EU); it will not occur until the appropriate legal structuring and authorizations are in place.